A nonlinear cavity constant and robust realized-start tests for critical SK slowing down

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M431

October 9, 2026

Abstract

We study the critical zero-field Gaussian Sherrington–Kirkpatrick heat-bath manuscript, keeping its Gibbs measure, Gaussian diagonal augmentation and two update clocks explicit. Conditional on its squared-overlap and covariance-scale inputs, we sharpen the linear inverse Rayleigh comparison to an asymptotic constant: $\mathcal{R}_{\mathrm{lin}}/\|\Sigma\| \to c_* = e^{1/2}/\mathbb{E}[\operatorname{sech} G]$, where $G \sim N(0,1)$ and $1 < c_* < e$. The new step controls both exact cavity factors rather than bounding one by Jensen alone. A clipped exceptional-site argument gives an additive $n^{1/3+\varepsilon}$ error at each fixed tolerance. Separately, we prove a finite sign-cell TV certificate with actual joint cell masses and Gibbs-metric direction errors. It transfers the source's realized-start obstruction to bounded even Hamiltonian perturbations, but does not infer independent projection signs, a full Poincare inequality or a new critical mixing exponent. All global spectral, spherical and orientation inputs remain identified as inherited. Contribution assessment: C3/5, the nonlinear cavity constant and quantitative robust-start extension are notable source-specific increments conditional on explicitly retained static inputs.

Contents

1 Model, clocks and the source mechanism **2**
1.1 The physical forms . . . 2
1.2 What is inherited . . . 3

2 Both cavity factors, not Jensen alone **3**

3 The asymptotic nonlinear cavity constant **4**

4 A finite multivariate realized-start certificate **6**
4.1 Covariance-aware approximate directions . . . 7

5 Worked examples and failure boundaries **8**
5.1 Two spins: the clocks and rare-disorder warning are exact . . . 8
5.2 A joint sign law cannot be supplied by orthogonal coefficients . . . 8
5.3 What a direction-error certificate must contain . . . 9

6 Original Ram comparison and contribution limits **9**

7 Public context and verification **10**

1 Model, clocks and the source mechanism

Let $\Omega_n = \{-1,1\}^n$, $n \geq 3$. The off-diagonal entries $A_{ij} = A_{ji}$, $i < j$, are independent $N(0, 1/n)$, and $A_{ii} = 0$. At inverse temperature one, with no external field,

$$H_A(x) = \sum_{i<j} A_{ij} x_i x_j, \qquad \mu_A(x) = \frac{e^{H_A(x)}}{\sum_y e^{H_A(y)}}.$$

Add an independent diagonal $D_{ii} \sim N(0, 2/n)$, and put $W = A + D$. Then W is GOE and $x^T W x/2 - H_A(x) = \operatorname{Tr} D/2$ is constant on the cube. The measure and heat-bath kernels are unchanged, but a top unit eigenvector $v = v(W)$ may depend on D.

For $P_i f = \mathbb{E}_\mu[f \mid x_{-i}]$, the exact clock relation is

$$\mathcal{L} = \sum_i (P_i - I), \qquad P = \frac{1}{n}\sum_i P_i, \qquad \mathcal{L} = n(P - I). \tag{1}$$

Continuous time has rate one *at every site*; one discrete step is one uniform-site attempt, which may leave the spin unchanged. Total variation means $\|\nu - \mu\|_{\mathrm{TV}} = \sup_B |\nu(B) - \mu(B)|$. The source's main claim [1] is, for every prescribed deterministic sequence,

$$t_n = o(n^{2/3}) \implies \mu_A\{x : \|e^{t_n \mathcal{L}}(x,\cdot) - \mu_A\|_{\mathrm{TV}} > 1/4\} \xrightarrow{\mathbb{P}_A} 1,$$
$$k_n = o(n^{5/3}) \implies \mu_A\{x : \|P^{k_n}(x,\cdot) - \mu_A\|_{\mathrm{TV}} > 1/4\} \xrightarrow{\mathbb{P}_A} 1.$$

The initial configuration is fixed before its transition law is tested. Averaging that law over μ_A would give equilibrium exactly and would erase the statement. No disorder-adaptive-time quantifier is silently added.

1.1 The physical forms

Spin-flip symmetry gives $\mu[x] = 0$, so

$$\Sigma = \mu[xx^T], \quad \Sigma_{ii} = 1, \quad Q_n = \mathbb{E}_{\mu^{\otimes 2}}(x^T y/n)^2, \quad \operatorname{Tr}\Sigma^2 = n^2 Q_n. \tag{2}$$

Conditional expectation is orthogonal projection in $L^2(\mu)$, not in counting measure. With $g_i = \sum_{j\neq i} A_{ij} x_j$ and $\alpha_i = \mu[\operatorname{sech}^2 g_i]$,

$$\mathcal{D}_\mu(f) = \sum_i \mu[\operatorname{Var}(f \mid x_{-i})], \qquad \mathcal{D}_\mu(a^T x) = \sum_i \alpha_i a_i^2. \tag{3}$$

Indeed, after conditioning only $a_i x_i$ remains random, with conditional variance $a_i^2 \operatorname{sech}^2 g_i$. Thus, writing $D_\alpha = \operatorname{diag}(\alpha_i)$,

$$\mathcal{R}_{\mathrm{lin}} = \sup_{a\neq 0} \frac{a^T \Sigma a}{a^T D_\alpha a} = \|D_\alpha^{-1/2} \Sigma D_\alpha^{-1/2}\|. \tag{4}$$

Full support makes every $\alpha_i > 0$. This is a linear-observable quantity, not the full inverse gap. Also, P has spectrum in $[0,1]$ because it is an average of orthogonal projections.

1.2 What is inherited

The full cached scientific text was freshly read. The following is a dependency map, not an independent certification of every global argument.

Source part	Mechanism and retained obligation
Spectral	GOE tridiagonalization; simultaneous block-prefix noise control; Hermite resolvent identity; an eigenvalue-only good event.
Spherical	Exact Gaussian radial disintegration; density lower bound at the mean; two-square smoothing after coordinate deletion.
Orientation	Haar first moment; cubic pair-overlap deficit; disjoint outward bins; relative partition concentration before division.
Dynamics	Stationary increments and a realized-start sign test; conditionally clipped cavity counts; separate high-temperature upper comparison.
Specification	A stationary measure alone does not specify a mixing time; rare disorder is not a high-probability asymptotic bound.

The inherited static statements used below are

$$Q_n = O_{\mathbb{P}}(n^{-2/3}), \tag{5}$$

$$\mu_A\{|v^T x| \le b_n n^{1/3}\} \xrightarrow{\mathbb{P}_{A,D}} 0 \quad \text{for every deterministic } b_n \downarrow 0, \tag{6}$$

$$\mathbb{P}\{n^{2/3}/M_n \le \|\Sigma\| \le M_n n^{2/3}\} \longrightarrow 1 \quad \text{for every deterministic } M_n \to \infty. \tag{7}$$

Monotonicity of b_n is unnecessary: any positive sequence tending to zero is allowed. The source derives (7) from the first two inputs: $\|\Sigma\|^2 \le n^2 Q_n$ supplies the upper bound, and $v^T\Sigma v \ge b_n^2 n^{2/3}\mu(|v^T x| > b_n n^{1/3})$ supplies the lower.

To make the scale visible, its spectral event uses $s_0 = L_n n^{-2/3}$, with $L_n \to \infty$ as slowly as required, $L_n \le \log n$. There $\lambda_1, \lambda_2, \lambda_3 = 2 + o(s_0)$ and two resolvent traces are controlled independently of orientation. At the spherical saddle, $k_i = (z - \lambda_i)^{-1}$, $\sum_i k_i = n$, $k_j \ge c/s_0$ for $j \le 3$, and

$$p(n) \ge c/(n\sqrt{s_0}), \qquad \sup p_{-i} \le C s_0.$$

These give a spherical small-window bound $CL_n^2 b_n$. Its transfer requires the restricted Haar first moment *and* concentration of the unrestricted partition sum. A covariance estimate alone does not supply anti-concentration. Choosing L_n after a deterministic threshold is essential; no effective universal rate in (6) is assumed.

2 Both cavity factors, not Jensen alone

Delete site i and its incident edges, without changing the $1/n$ variance of the remaining couplings. Let ν_i be this $(n-1)$-spin law, $\mathcal{F}_i$ its disorder sigma-field, and set

$$Z_i = \nu_i[\cosh g_i], \quad Y_i = \nu_i[\operatorname{sech} g_i], \quad R_i = \mathbb{E}_{\nu_i^{\otimes 2}}(x_{-i}^T y_{-i}/n)^2. \tag{8}$$

Summing over x_i gives the exact identities

$$\frac{d\bar{\mu}_i}{d\nu_i} = \frac{\cosh g_i}{Z_i}, \qquad \alpha_i = Y_i/Z_i, \qquad \alpha_i^{-1} \le Z_i^2. \tag{9}$$

The last inequality is Jensen applied to $u \mapsto 1/u$, but replacing Y_i by $1/Z_i$ loses useful information. We retain its actual Gaussian expectation.

Lemma 2.1 (Even Gaussian covariance)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be bounded, even and twice continuously differentiable, with bounded first derivative and $\|f''\|_\infty \le K$. If (X, Y) is centered Gaussian with variances $s > 0$ and covariance $c \in [-s, s]$, then*

$$|\operatorname{Cov}(f(X), f(Y))| \le \tfrac{1}{2} K^2 c^2. \tag{10}$$

In particular this holds for $f = \operatorname{sech}$ with $K = 1$.

Proof. For $|c| < s$, put $F(c) = \mathbb{E}[f(X)f(Y)]$. The bivariate Gaussian density satisfies $\partial_c p_c = \partial_x \partial_y p_c$. Differentiating on compact subintervals of $(-s, s)$ and integrating by parts gives $F'(c) = \mathbb{E}[f'(X)f'(Y)]$ and $F''(c) = \mathbb{E}[f''(X)f''(Y)]$. Gaussian tails and the derivative bounds justify the operations; alternatively first use compact cutoffs and pass by domination. At zero the variables are independent, and evenness gives $F'(0) = (\mathbb{E} f'(\sqrt{s}G))^2 = 0$. Taylor's integral remainder and $|F''| \le K^2$ prove the claim in the open interval. Bounded continuity extends it to the degenerate endpoints. For $u = \operatorname{sech} x \in (0, 1]$, $f'' = u - 2u^3$, whose absolute value is at most one. This is the classical Gaussian interpolation identity, used here for a specific cavity factor, not claimed as a new identity. □

Write $s_n = (n-1)/n$, and for $G \sim N(0, 1)$ define

$$z_n = e^{s_n/2}, \qquad y_n = \mathbb{E}[\operatorname{sech}(\sqrt{s_n} G)], \qquad c_n = z_n / y_n, \qquad c_* = \frac{e^{1/2}}{\mathbb{E}[\operatorname{sech} G]}. \tag{11}$$

These are deterministic Gaussian integrals, not expectations under counting measure on the cube. Dominated convergence gives $c_n \to c_*$. Strict Jensen yields $\mathbb{E} \operatorname{sech} G > 1/\mathbb{E} \cosh G = e^{-1/2}$, and $\mathbb{E} \operatorname{sech} G < 1$, so $1 < c_* < e$.

Proposition 2.2 (Conditional two-factor concentration)**.** *For every site, conditional on $\mathcal{F}_i$,*

$$\mathbb{E}[Z_i \mid \mathcal{F}_i] = z_n, \qquad \operatorname{Var}(Z_i \mid \mathcal{F}_i) \le C R_i, \tag{12}$$

$$\mathbb{E}[Y_i \mid \mathcal{F}_i] = y_n, \qquad \operatorname{Var}(Y_i \mid \mathcal{F}_i) \le R_i/2. \tag{13}$$

For every fixed integer $m \ge 1$, $\mathbb{E}[Z_i^m \mid \mathcal{F}_i] \le 2e^{m^2/2}$. Also, deterministically,

$$R_i \le 2 Z_i^2 (Q_n + n^{-2}). \tag{14}$$

Proof. The incident row is independent of $\mathcal{F}_i$. For each fixed configuration its local field is $N(0, s_n)$; for two configurations the covariance is $c = x_{-i}^T y_{-i}/n$. Their Gaussian cosh-product mean is $e^{s_n} \cosh c$, so $\operatorname{Var}(Z_i \mid \mathcal{F}_i) = e^{s_n} \nu_i^{\otimes 2}[\cosh c - 1] \le C R_i$. Apply Lemma 2.1 inside the double expectation to obtain the second variance bound. Interchanging the finite spin sums and Gaussian integrals is legitimate. Jensen and $\mathbb{E} \cosh^m(\sqrt{s_n} G) \le 2e^{m^2/2}$ prove the moment bound. Finally $d\nu_i / d\bar{\mu}_i = Z_i / \cosh g_i \le Z_i$; taking two replicas and using $(x_{-i}^T y_{-i}/n)^2 \le 2(x^T y/n)^2 + 2n^{-2}$ proves (14). No conditioning on a multi-row good event is used. □

3 The asymptotic nonlinear cavity constant

Theorem 3.1 (Sharper linear comparison)**.** *Assume the source overlap input (5) for critical Gaussian SK disorder. For every fixed $0 < \varepsilon < 1/3$ and $0 < \eta < c_*$, with probability tending to one,*

$$(c_* - \eta)\|\Sigma\| - n^{1/3+\varepsilon} \le \mathcal{R}_{\mathrm{lin}} \le (c_* + \eta)\|\Sigma\| + n^{1/3+\varepsilon}. \tag{15}$$

If also the covariance lower-scale input (7) holds, then

$$\frac{\mathcal{R}_{\mathrm{lin}}}{\|\Sigma\|} \xrightarrow{\mathbb{P}} c_*. \tag{16}$$

These statements concern only linear tests; they do not identify the full relaxation constant.

Proof. Fix the two tolerances first. Since $y_n \to y_* := \mathbb{E}\operatorname{sech} G > 0$ and $z_n \to z_* := e^{1/2}$, choose a fixed $u > 0$ small enough that, for all large n,

$$|Z - z_n| \le u, \quad |Y - y_n| \le u \quad \Longrightarrow \quad c_* - \eta \le Z/Y \le c_* + \eta. \tag{17}$$

For example, first ensure $u < y_*/4$, and then use continuity of the ratio on a rectangle bounded away from $Y = 0$.

Set $a = \varepsilon/8$, and choose a fixed integer $m > 1/a$. The moment bound and a union bound give

$$\mathbb{P}(\max_i Z_i > n^a) \le 2e^{m^2/2} n^{1-am} = o(1).$$

Tightness in (5) implies $\mathbb{P}(Q_n > n^{-2/3+a}) = o(1)$. On the intersection of their complements, (14) gives, for all large n,

$$\max_i R_i \le r_n := 3n^{-2/3+3a}.$$

Let B be the set where either inequality on the left of (17) fails. Define the clipped count

$$N_n = \sum_i \mathbf{1}_{\{R_i \le r_n\}} \mathbf{1}_{\{|Z_i - z_n| > u \text{ or } |Y_i - y_n| > u\}}.$$

Because R_i is $\mathcal{F}_i$-measurable, conditional Chebyshev and Proposition 2.2 give

$$\mathbb{E}N_n \le C_u n r_n \le C'_u n^{1/3+3a}, \qquad \mathbb{P}(N_n > n^{1/3+4a}) \le C'_u n^{-a}.$$

On $\max_i R_i \le r_n$, $N_n = |B|$. Thus the event

$$\max_i Z_i \le n^a, \qquad |B| \le n^{1/3+4a} \tag{18}$$

has probability tending to one. More explicitly, its failure probability is at most

$$\mathbb{P}(Q_n > n^{-2/3+a}) + 2e^{m^2/2} n^{1-am} + C'_u n^{-a}$$

for sufficiently large n. No independence between exceptional sites is asserted, and the Gaussian calculation was never conditioned on the global event (18).

Write $G = B^c$ and $\Gamma = D_\alpha^{-1/2} \Sigma D_\alpha^{-1/2}$. For a positive semidefinite block matrix K,

$$\|K\| \le \|K_{GG}\| + \|K_{BB}\|. \tag{19}$$

Indeed, positivity gives $|x^T K_{GB} y|^2 \le (x^T K_{GG} x)(y^T K_{BB} y)$, and Cauchy–Schwarz bounds the full form by the stated sum times $\|x\|^2 + \|y\|^2$. On G, (9) and (17) put every α_i^{-1} in $[c_* - \eta, c_* + \eta]$, whence

$$(c_* - \eta)\|\Sigma_{GG}\| \le \|\Gamma_{GG}\| \le (c_* + \eta)\|\Sigma_{GG}\|.$$

This is a statement about largest eigenvalues of diagonal congruences, not an invalid entrywise Loewner comparison. The lower bound follows by testing at $D^{-1}v$, where v is a unit top eigenvector of Σ_{GG} and $D = \operatorname{diag}(\alpha_i^{-1/2})_{i \in G}$. The upper bound follows from operator norms.

The bad block has

$$\|\Gamma_{BB}\| \le \operatorname{Tr}\Gamma_{BB} = \sum_{i\in B} \alpha_i^{-1} \le |B| \max_i Z_i^2 \le n^{1/3+6a}.$$

Also $\|\Sigma_{BB}\| \le |B|$, so (19) implies $\|\Sigma_{GG}\| \ge \|\Sigma\| - |B|$. The same formulas cover an empty block, with its norm zero. Consequently

$$(c_* - \eta)(\|\Sigma\| - n^{1/3+4a}) \le \|\Gamma\| \le (c_* + \eta)\|\Sigma\| + n^{1/3+6a}.$$

Since $6a < \varepsilon$, all fixed constants are absorbed into the displayed error of (15) for large n.

Finally choose $d = (1/3 - \varepsilon)/2 > 0$. Input (7) with $M_n = n^d$ gives $\|\Sigma\| \ge n^{2/3-d}$ with high probability. Thus $n^{1/3+\varepsilon}/\|\Sigma\| \le n^{-d}$ there. Divide (15) by $\|\Sigma\|$, then let the fixed tolerance η be arbitrarily small. This proves (16). □

The source's displayed comparison is $\|\Sigma\| \le \mathcal{R}_{\text{lin}} \le 16\|\Sigma\| + n^{13/25}$. Optimizing its threshold $Z_i > 4$ alone would improve the coefficient to any number above e, with a smaller exceptional-block exponent. Theorem 3.1 additionally retains Y_i, improves that coefficient to $c_* < e$, and obtains a two-sided asymptotic ratio. The mean $\mathbb{E}\operatorname{sech} G$ is essential: substituting a reciprocal mean would undo the improvement. The assertion is conditional on the stated critical static inputs, not a replacement for their proofs or a historical priority claim about cavity theory.

4 A finite multivariate realized-start certificate

Here the finite theorem is independent of GOE. Let ρ be any full-support probability on Ω_n, let $c_i > 0$ be fixed site rates, and write

$$\mathcal{L}_{\rho,c} = \sum_i c_i(P_i^\rho - I), \qquad C_\Sigma = \sum_i c_i, \qquad K_{\rho,c} = \sum_i (c_i/C_\Sigma) P_i^\rho.$$

Then $\mathcal{L}_{\rho,c} = C_\Sigma(K_{\rho,c} - I)$. Rates may depend on the fixed disorder, but not on the running configuration. Uniform attempts have $c_i = 1$ and $C_\Sigma = n$. For either $T = e^{t\mathcal{L}_{\rho,c}}$ or $T = K_{\rho,c}^k$, define $\tau = t$ or $\tau = k/C_\Sigma$, respectively. Set

$$\mathcal{D}_{\rho,c}(f) = \sum_i c_i \rho[\operatorname{Var}_\rho(f \mid x_{-i})].$$

Exactly as in the source, reversibility and the spectral inequalities $1 - e^{-t\lambda} \le t\lambda$, $1 - \theta^k \le k(1-\theta)$ for $0 \le \theta \le 1$ imply

$$\mathbb{E}_\rho[(f(X') - f(X))^2] \le 2\tau \mathcal{D}_{\rho,c}(f), \qquad X \sim \rho, \quad X' \mid X \sim T(X, \cdot). \tag{20}$$

Choose real functions $f_1, \dots, f_r$, and margins $a_j > 0$. For $s \in \{-1, 1\}^r$, set

$$C_s = \{x : s_j f_j(x) > 0 \text{ for every } j\}, \quad p_s = \rho(C_s), \quad Z = \{x : \text{some } f_j(x) = 0\}.$$

For $x \notin Z$, let $s(x)$ be its sign label and $R(x) = 1 - T(x, C_{s(x)})$; set $R(x) = 1$ on Z. Define the explicit error budget

$$B = \sum_{j=1}^r \rho\{|f_j| \le a_j\} + 2\tau \sum_{j=1}^r \frac{\mathcal{D}_{\rho,c}(f_j)}{a_j^2}. \tag{21}$$

Theorem 4.1 (Joint-cell test)**.** *One has $\rho[R] \le B$. For $x \notin Z$,*

$$\|T(x,\cdot) - \rho\|_{\mathrm{TV}} \ge 1 - p_{s(x)} - R(x). \tag{22}$$

Consequently, for $u \in [0,1]$ and $\zeta > 0$,

$$\rho\{x : \|T(x,\cdot) - \rho\|_{\mathrm{TV}} \le 1 - u - \zeta\} \le \sum_{s:p_s>u} p_s + B/\zeta. \tag{23}$$

No independence or product approximation for the r signs is assumed.

Proof. If $|f_j(X)| > a_j$, changing its sign or reaching zero requires $|f_j(X') - f_j(X)| \ge a_j$. The union bound, Markov's inequality and (20) give the bound for $\rho[R]$, including starts in Z. Testing TV against the cell of the fixed start gives (22). Outside Z, if $p_{s(x)} \le u$ and $R(x) < \zeta$, this lower bound is strictly above $1 - u - \zeta$. Zeros are already contained in $\{R \ge \zeta\}$ when $\zeta \le 1$. Thus Markov gives (23); for $\zeta > 1$, its left side is zero. Empty cells cause no problem. □

For a spin-flip symmetric law and one nonzero odd observable, $p_+, p_- \le 1/2$. Taking $u = 1/2$, $\zeta = 1/4$ recovers the source's bad-start bound $4B$. A bank can improve the TV threshold only if its *joint* cell masses are small. For example, repeating one observable r times still leaves two cells of mass at most one half, not 2^r cells of mass 2^{-r}. A coordinate transformation does not establish a new sign law.

4.1 Covariance-aware approximate directions

Let μ be even, let $\rho = e^V\mu/\mu[e^V]$ with $V(-x) = V(x)$ and $\operatorname{osc} V \le R_0$, and consider $f_j(x) = w_j^T x$ approximating $v_j^T x$. Define the *full* Gibbs error

$$e_j^2 = (w_j - v_j)^T \Sigma_\mu (w_j - v_j) = \mu[((w_j - v_j)^T x)^2].$$

For any additional tolerances $b_j > 0$,

$$\rho\{|w_j^T x| \le a_j\} \le e^{R_0}\left(\mu\{|v_j^T x| \le a_j + b_j\} + e_j^2/b_j^2\right). \tag{24}$$

Indeed, outside $\{|(w_j - v_j)^T x| > b_j\}$ the event on the left implies the small-window event on the right. Apply Markov in $L^2(\mu)$ and $d\rho/d\mu \le e^{R_0}$. The new heat-bath form, not the old kernel, must be used:

$$\mathcal{D}_{\rho,c}(w_j^T x) = \sum_i c_i w_{ji}^2 \rho[\mathrm{Var}_\rho(x_i \mid x_{-i})] \le \sum_i c_i w_{ji}^2. \tag{25}$$

Thus a valid replacement for B in Theorem 4.1 is

$$B_{\mathrm{cert}} = e^{R_0}\sum_j \left(\mu\{|v_j^T x| \le a_j + b_j\} + \frac{e_j^2}{b_j^2}\right) + 2\tau \sum_j \frac{\sum_i c_i w_{ji}^2}{a_j^2}. \tag{26}$$

Use the actual new cell masses under ρ in (23). This separates equilibrium distortion, direction error and dynamic motion. The factor $n\|w_j - v_j\|^2$ is a valid but potentially wasteful bound on e_j^2. A certified upper bound on the physical quadratic form can replace it. A few sampled moments or a projected residual cannot.

Corollary 4.2 (Bounded even perturbations preserve the obstruction)**.** *Assume* (6)*. Let* $V_n = V_n(A, x)$ *be measurable, even in* x*, with* $\operatorname{osc}_x V_n \le R_0 < \infty$ *for every disorder and dimension. Put* $\rho_A \propto e^{V_n(A,\cdot)}\mu_A$*. For continuous heat-bath dynamics for* ρ_A*, allow positive disorder-dependent site rates* $c_i(A) \le C < \infty$*. For every deterministic* $t_n = o(n^{2/3})$*,*

$$\rho_A\{x : \|e^{t_n\mathcal{L}_{\rho,c}}(x,\cdot) - \rho_A\|_{\mathrm{TV}} > 1/4\} \xrightarrow{\mathbb{P}_A} 1.$$

For the new uniform-site *heat-bath kernel the analogous conclusion holds at every deterministic integer* $k_n = o(n^{5/3})$*.*

Proof. For each (A, D) use the source top unit vector $v(A + D)$, with no approximation error. Choose deterministic $b_n \to 0$ with $t_n/(b_n^2 n^{2/3}) \to 0$, and take margin $a_n = b_n n^{1/3}$. The error in (21) is at most

$$e^{R_0}\mu_A\{|v^T x| \le a_n\} + 2Ct_n/a_n^2.$$

Its expectation over (A, D) tends to zero: the first mass is bounded by one and converges in probability, and the second term is deterministic. Spin-flip symmetry of ρ_A gives the half-space mass bound. For each fixed start the intrinsic bad-state indicator is at most $4R(x)$. Integrate *this indicator* over D, then over A, and apply Markov to its ρ_A-mass. The law ρ_A and its dynamics do not depend on the auxiliary diagonal, so the resulting disorder statement is over A alone. No transition law is averaged in place of its distance. For uniform attempts, (20) replaces t_n by k_n/n; the same choice works on the scale $n^{5/3}$. □

The corollary permits bounded many-spin even potentials, not just a small matrix perturbation. For a symmetric coupling error E_n, $V_n(x) = x^T E_n x/2$ satisfies $\operatorname{osc} V_n \le n\|E_n\|$, so $n\|E_n\| = O(1)$ is one sufficient condition; a diagonal error costs nothing. This condition is not necessary. Unbounded $R_0 = R_n$ requires the explicit first term in (26) to vanish, which does not follow from qualitative anti-concentration alone. External fields break the half-space symmetry; Theorem 4.1 remains valid but its actual cell masses must be used. The corollary gives no all-function gap or matching mixing upper bound.

5 Worked examples and failure boundaries

5.1 Two spins: the clocks and rare-disorder warning are exact

Take $\mu(x) \propto e^{wx_1x_2}$, $r = \tanh w \in [0, 1)$. Then

$$\Sigma = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad \alpha_1 = \alpha_2 = 1 - r^2, \qquad \|\Sigma\| = 1 + r, \qquad \mathcal{R}_{\mathrm{lin}} = \frac{1}{1-r}.$$

For $f = x_1 + x_2$, $Pf = (1+r)f/2$ and $\mathcal{L}f = -(1-r)f$. At $e^w = 2$, $r = 3/5$, these values are $\alpha = 16/25$, $\|\Sigma\| = 8/5$, $\mathcal{R}_{\mathrm{lin}} = 5/2$, and the discrete eigenvalue is $4/5$. The ratio $\mathcal{R}_{\mathrm{lin}}/\|\Sigma\| = 1/(1 - r^2)$ can diverge as $w \to \infty$. Thus Theorem 3.1 is neither a fixed-dimension identity nor an essential bound over every disorder. This example rederives the source's clock warning rather than claiming a new low-dimensional mixing theorem.

5.2 A joint sign law cannot be supplied by orthogonal coefficients

With the same two-spin law and $f_1 = x_1, f_2 = x_2$, each sign is exactly balanced, but

$$p_{++} = p_{--} = (1 + r)/4, \qquad p_{+-} = p_{-+} = (1 - r)/4.$$

The coefficient vectors are Euclidean orthogonal. At $r = 3/5$ the largest cell has mass $2/5$, not $1/4$. The zero-motion level of the cell test is therefore $3/5$, not $3/4$. These coordinate tests have order-one Dirichlet costs and do not create a critical slow mode. In general an enlarged bank is useful only after its margins, Dirichlet costs and joint Gibbs masses have all been controlled.

5.3 What a direction-error certificate must contain

For an even law, if $e^2 = (w - v)^T \Sigma (w - v)$, one approximate direction with margins a, b gives

$$B_{\mathrm{cert}} \le e^{R_0} \mu(|v^T x| \le a + b) + e^{R_0} e^2 / b^2 + 2\tau C \|w\|^2 / a^2.$$

All three terms have different origins. In the critical regime the natural magnitude is $n^{1/3}$, so an error $e = o(n^{1/3})$ can be accommodated by thresholds tending to zero on that scale, provided the motion term is also small. This is a conditional criterion, not an assertion that an arbitrary numerical top eigenvector meets it. The source covariance upper bound turns a deterministic Euclidean error $\|w - v\| \to 0$ into $e/n^{1/3} \to 0$ in probability when $n^{-2/3}\|\Sigma\|$ is tight; a quantitative certificate still requires a specified upper bound for the realized error.

6 Original Ram comparison and contribution limits

The original physical-metric chapter [6], freshly read in full, derives $G = J^* J$, the physical adjoint and an orthogonal decomposition of the full residual into retained and omitted parts. Here the synthesis is $Ja = a^T x$ into $L^2(\mu)$, so $G = \Sigma$, while the Dirichlet form has the separate matrix D_α. An invertible coefficient change transports both by congruence. It does not replace Σ by the identity, change α_i to one, or prove invariance of the linear-observable space under the heat-bath generator. In fact $\mathcal{L}(a^T x) = \sum_i a_i(\tanh g_i - x_i)$ is generally nonlinear in the spins.

Fresh original TR250 passages [7] give the physical-space restriction, energy comparison and the warning that a finite probe norm is only a lower bound on a full dual residual. We use that discipline in (24), which needs the whole Gibbs quadratic error. No PDE theorem, unobserved residual estimate, or prior comparative report is imported as an SK spectral theorem. These original comparisons explain how to preserve the norm; the new content is the two-factor cavity calculation and the explicit dynamic robustness consequences, not the coordinate change itself.

Advantages. The constant c_* records the nonlinear one-site conditional variance under the correct cavity reweighting. The exceptional-block argument requires no independence between sites and uses only the source overlap scale. The TV certificate allows several observables, finite numerical direction errors, and genuinely changed even Gibbs laws while keeping their kernels and clock normalization explicit.

Costs and limitations. Neither a full Gibbs covariance nor the joint cell masses are free computational inputs; direct enumeration costs exponentially many spin states. No efficient oracle or measured speedup is claimed. The leading constant is an exact Gaussian integral, not a fitted numerical value. The ratio limit still depends on high-probability critical static inputs. Its finite-size constants depend on the chosen tolerances, and it does not provide an optimal remainder. Bounded energy tilts need not be small in total variation, but unbounded tilts and odd perturbations require additional estimates. The source's all-function subexponential upper bound imports a separate fixed-subcritical-temperature Poincare theorem; neither our linear comparison nor our lower TV certificate supplies it.

Further work. Quantitative versions of the overlap input would give explicit confidence bounds in the clipped-count proof. A joint law for several edge projections could make the cell test substantially stronger than a half-space test, but must be proved rather than inferred from marginal symmetry. Validated Gaussian quadrature could enclose c_*; it would improve numerical reporting, not the theorem. Extending the perturbation corollary to growing energy oscillation needs quantitative small-window control before taking the dimension limit.

7 Public context and verification

Primary public records were checked on October 9, 2026. Du–Huang's July 8 version [2] states critical free-energy fluctuation and overlap-scale results; their August 9 overlap preprint [3] states an explicit random quenched limiting overlap law through the GOE edge. These are relevant static inputs, not substitutes for specifying heat-bath clocks and fixed-start transition laws. Wang's August 28 version [4] treats fixed $\beta < 1/2$, including external fields, and Boban–Li–Oveis Gharan's September 11 preprint [5] extends a polynomial-mixing regime above one half. Neither abstract establishes the present critical realized-start result. Only primary abstracts and metadata of these external papers were freshly inspected; their complete proofs remain outside this report's read scope.

The owned ledger records all eleven cached public text files, the full 210-line original physical-metric chapter, and the displayed original TR250 intervals. Included but unopened private references and the metric figure are excluded. Pre-read hashes were saved before these literal reads and compared with post-read hashes. Captured-October-8 source authentication is provenance, not a live-branch or global-proof verdict.

The independently executable exact-check script uses rational arithmetic for finite Gibbs deletion identities, clocks, cell tests, direction-error bounds, positive block comparisons and exponent bookkeeping. Polynomial Gaussian moment fixtures check the interpolation constant; rational sech-derivative fixtures check its stated bound. Such cases are not a numerical evaluation of $\mathbb{E}\,\operatorname{sech} G$, a random-disorder experiment or a substitute for the asymptotic proof. Parent checks are separate and excluded from the author totals recorded in the final receipt.

The contribution score is C3, independently of applicability L3. The nonlinear ratio limit is a specific extension of the source's conservative linear comparison, and the perturbation theorem is a proved conditional extension of its actual sign-test mechanism. No source correction is asserted, no matching critical mixing theorem is independently claimed, and classical Gaussian interpolation and Hilbert-space inequalities receive no invented novelty credit.

References

[1] OpenAI. *Critical slowing down in the Sherrington–Kirkpatrick model.* September 24, 2026. Pinned public manuscript.

[2] H. Du and B. Huang. *Fluctuations of the Sherrington–Kirkpatrick free energy at critical temperature.* arXiv:2607.02172v2, July 8, 2026.

[3] H. Du and B. Huang. *Overlap distribution of the critical Sherrington–Kirkpatrick model.* arXiv:2608.08752v1, August 9, 2026.

[4] S. Wang. *Optimal Mixing of Glauber Dynamics for the Sherrington–Kirkpatrick Model at $\beta < 1/2$.* arXiv:2608.22159v2, August 28, 2026.

[5] M. Boban, A. Li and S. Oveis Gharan. *Rank-1-perturbed trickledown theorems: Mixing time of Glauber dynamics for the Sherrington–Kirkpatrick model up to* $\beta \leq 1/2 + \varepsilon$. arXiv:2609.13138v1, September 11, 2026.

[6] Private Ram corpus. Original physical-metric chapter, volume 21, lines 1–210 freshly read. Confidential local comparison.

[7] Private Ram corpus. Original TR250, lines 121–181, 425–477, 564–599 and 1124–1160 freshly read. Confidential local proof comparison; other parts and cited originals not claimed read.